PECH GROUP HOLDINGS LTD

Technology & Infrastructure Enablers for Africa

Open-Source Platform Stack — Complete Reference

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1. LICENSE POLICY

PECH only uses platform tools with **Apache-2.0, MIT, or BSD** licenses. This allows us to: - Modify and extend without sharing source code - Embed in proprietary products - Rebrand under PECH identity - Sell as part of subscription services

Excluded license types: AGPL-3.0, GPL-3.0 (for platforms we modify and serve), Sustainable Use License, SSPL, BSL (Business Source License).

2. IDENTITY & ACCESS MANAGEMENT

Primary: Keycloak

Attribute	Detail
Website	keycloak.org
GitHub	github.com/keycloak/keycloak
License	Apache-2.0
Language	Java (Quarkus)
PECH Use	SSO across all 10 verticals, OAuth2/OIDC, user federation, social login, RBAC, multi-tenancy

Why Keycloak: - Industry standard for enterprise IAM - Supports social login (Google, Facebook, Apple) - Multi-realm for PECH's multi-tenant architecture (marketplace vendors, fintech agents, solar installers each get their own realm) - Self-hosted, no per-user fees

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Authentik | Custom (source-available) | Less mature; enterprise features require paid license | | Zitadel | Apache-2.0 | Newer; smaller community; could be future option | | SuperTokens | Apache-2.0 | Good for simple auth but lacks Keycloak's multi-realm enterprise features |

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3. API GATEWAY & SERVICE MESH

Primary: Apache APISIX

Attribute	Detail
Website	apisix.apache.org
GitHub	github.com/apache/apisix
License	Apache-2.0
Language	Lua (OpenResty/Nginx)
PECH Use	API gateway for all 78 microservices, rate limiting, auth, load balancing, circuit breaking

Why APISIX: - Highest performance open-source API gateway (benchmarks show 2-5× faster than Kong) - Plugin system for custom logic (auth, rate limiting, AI model routing) - Dashboard for visual management - Dynamic upstream configuration (route AI requests to different model servers)

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Kong | Apache-2.0 (Community) | Heavier; enterprise features are proprietary; APISIX is faster | | Traefik | MIT | Better for simple setups; lacks APISIX's plugin ecosystem | | Envoy | Apache-2.0 | Lower-level; better as sidecar proxy than standalone gateway |

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4. E-COMMERCE / MARKETPLACE

Primary: Medusa.js

Attribute	Detail
Website	medusajs.com
GitHub	github.com/medusajs/medusa
License	MIT
Language	TypeScript (Node.js)

PECH Use	Multi-vendor marketplace, product catalog, cart, checkout, order management, vendor storefronts

Why Medusa: - Headless architecture — PECH builds its own frontend - MIT license — can fully customize and close-source - Plugin system for multi-vendor extension - Modern stack (TypeScript, PostgreSQL) - Built-in payment and fulfillment abstractions

Medusa Extensions for PECH: - Multi-vendor plugin (vendor onboarding, commission splits) - Solar product catalog (panels, inverters, batteries) - Real estate listings (extend product model for property) - Installer marketplace (service products)

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Saleor | BSD-3 | Strong alternative; Django/Python stack; smaller plugin ecosystem | | WooCommerce | GPL-2.0 | WordPress dependency; PHP; license issues for PECH modifications | | Sharetribe Go | Source-available | Not truly open-source; Sharetribe Flex is proprietary SaaS |

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5. ERP / ACCOUNTING

Primary: ERPNext

Attribute	Detail
Website	erpnext.com
GitHub	github.com/frappe/erpnext
License	MIT (core Frappe framework) / GPL-3.0 (some ERPNext modules)
Language	Python (Frappe framework)
PECH Use	Accounting, HR, payroll, inventory, procurement, tax compliance (FIRS), auditing

Why ERPNext: - Full ERP suite in one platform - Strong accounting module (multi-currency, multi-company) - Nigerian tax compliance possible with customization - Python/Frappe makes AI integration straightforward - Large community in Africa

License Note: ERPNext's core Frappe framework is MIT. Some ERPNext modules are GPL-3.0. For PECH's accounting AI subscription product, we build our own AI layer on top (which is our proprietary code) and use ERPNext's MIT-licensed Frappe APIs. The GPL modules are used as-is without modification.

PECH AI Integration: - AI-powered bookkeeping (auto-categorize transactions) - Invoice OCR → automatic journal entries - Tax computation AI (FIRS, VAT, WHT) - Audit trail analysis and anomaly detection - Financial forecasting with Prophet/LightGBM

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Odoo Community | LGPL-3.0 | LGPL is OK but Odoo's ecosystem pushes toward Enterprise (proprietary) | | Apache OFBiz | Apache-2.0 | Outdated UI; smaller community; Java-heavy | | Akaunting | AGPL-3.0 | AGPL — cannot close-source modifications |

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6. FINTECH / CORE BANKING

Primary: Apache Fineract

Attribute	Detail
Website	fineract.apache.org
GitHub	github.com/apache/fineract

License	Apache-2.0
Language	Java (Spring Boot)
PECH Use	Wallet management, savings, loans, agent banking, transaction ledger, payment gateway integration

Why Fineract: - Built for financial inclusion in emerging markets - Proven in 100+ deployments across Africa and Asia - Core banking features: accounts, transactions, interest, fees - Agent banking module (perfect for PECH fintech agent network) - Apache-2.0 — full freedom to modify and rebrand

PECH Fintech Features: - PECH Wallet (powered by Fineract ledger) - Agent network management (cash-in/cash-out) - Micro-loans for solar equipment - Bill payments integration - Paystack/Flutterwave payment gateway integration

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Mifos X | Apache-2.0 | Same codebase as Fineract (Mifos donated to Apache) | | Cyclos | AGPL-3.0 | AGPL — cannot close-source | | Teller | Proprietary API | Not self-hosted |

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7. LOGISTICS & ROUTING

Primary: OSRM (Open Source Routing Machine)

Attribute	Detail
Website	project-osrm.org
GitHub	github.com/Project-OSRM/osrm-backend
License	BSD-2-Clause
Language	C++
PECH Use	Delivery route optimization, hub-and-spoke logistics, installer dispatch, real-time routing

Why OSRM: - Fastest open-source routing engine - Uses OpenStreetMap data (good Nigeria coverage from local mapping community) - Handles matrix routing (100+ delivery points optimization) - Self-hosted — no API call limits or costs

PECH Logistics Features: - Hub-and-spoke delivery optimization (Lagos hubs → distribution points) - Installer dispatch (nearest qualified installer) - Real-time delivery tracking - Fleet management route planning - Last-mile delivery optimization

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Valhalla | MIT | Slightly slower; less mature than OSRM | | GraphHopper | Apache-2.0 | Java-based; good alternative but OSRM is faster for pure routing | | Google Maps API | Proprietary | Expensive at scale; vendor lock-in |

Complementary Tools: | Tool | License | Purpose | |------|---------|---------| | OpenStreetMap | ODbL | Map data source for Nigeria | | Nominatim | GPL-2.0 | Geocoding (address → coordinates) — use as service, don't modify | | Leaflet.js | BSD-2 | Map frontend rendering |

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8. IoT PLATFORM

Primary: ThingsBoard Community Edition

Attribute	Detail
Website	thingsboard.io

GitHub	github.com/thingsboard/thingsboard
License	Apache-2.0 (Community Edition)
Language	Java
PECH Use	Solar panel monitoring, inverter telemetry, energy usage dashboards, device management, alerts

Why ThingsBoard: - Purpose-built for IoT device management - Real-time dashboards and rule engine - MQTT, CoAP, HTTP device protocols - Multi-tenant — each customer sees their own devices - Apache-2.0 — can extend and rebrand

PECH IoT Features: - Solar panel performance monitoring (voltage, current, temperature, yield) - Inverter status and fault alerts - Battery state-of-charge tracking - Energy consumption analytics - Predictive maintenance alerts (AI-powered) - Remote device configuration

> **Important:** Use **Community Edition only** (Apache-2.0). Professional Edition (PE) is proprietary.

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Mainflux (now SuperMQ) | Apache-2.0 | Lighter but less mature dashboards | | Node-RED | Apache-2.0 | Good for prototyping; not a full IoT platform | | AWS IoT Core | Proprietary | Vendor lock-in; expensive at scale |

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9. CUSTOMER SUPPORT

Primary: Chatwoot

Attribute	Detail
Website	chatwoot.com
GitHub	github.com/chatwoot/chatwoot
License	MIT
Language	Ruby on Rails
PECH Use	Customer support across all channels (web chat, WhatsApp, email, social), AI chatbot integration

Why Chatwoot: - Omnichannel support (web, WhatsApp, Facebook, email, Telegram) - WhatsApp integration is critical for Nigeria (most-used messaging app) - MIT license — embed and rebrand freely - API for AI chatbot integration (connect PECH's LLMs) - Conversation handoff from bot to human agents

PECH AI Integration: - First-response AI chatbot (Qwen2.5-7B answers common questions) - Auto-categorize and route tickets (DistilBERT) - Suggested responses for human agents - Sentiment analysis on conversations - Knowledge base search (RAG pipeline)

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Zammad | AGPL-3.0 | AGPL — cannot close-source modifications | | Rocket.Chat | MIT | More chat-focused; less support-workflow tooling | | Freshdesk | Proprietary | Per-agent pricing; vendor lock-in |

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10. WORKFLOW AUTOMATION

Primary: Apache Airflow

Attribute	Detail
Website	airflow.apache.org

GitHub	github.com/apache/airflow
License	Apache-2.0
Language	Python
PECH Use	Data pipelines, ETL, model training schedules, report generation, batch processing

Why Airflow: - Industry standard for workflow orchestration - Python-native — same language as PECH's AI/ML stack - DAG-based scheduling (define complex dependencies) - Huge connector ecosystem (databases, APIs, cloud services) - Apache-2.0 — no restrictions

PECH Workflows: - Daily RAG data ingestion (business data → embeddings → Qdrant) - Model retraining pipelines (weekly/monthly) - Financial report generation - Invoice processing pipeline (OCR → categorization → journal entry) - IoT data aggregation and alerting

> **Replaced n8n** — n8n uses a "Sustainable Use License" that restricts competing SaaS offerings. Since PECH may sell automation features, Airflow is the safe choice.

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | n8n | Sustainable Use License | **Cannot resell as automation platform** — license explicitly restricts this | | Prefect | Apache-2.0 | Good alternative; newer; consider for simpler workflows | | Temporal | MIT | Better for long-running workflows; more complex setup | | Dagster | Apache-2.0 | Modern alternative; smaller community than Airflow |

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11. BUSINESS INTELLIGENCE & ANALYTICS

Primary: Apache Superset

Attribute	Detail
Website	superset.apache.org
GitHub	github.com/apache/superset
License	Apache-2.0
Language	Python (Flask) + TypeScript (React)
PECH Use	Business dashboards, sales analytics, energy monitoring charts, financial reports, AI metrics

Why Superset: - Enterprise-grade BI with beautiful visualizations - SQL Lab for ad-hoc queries - Dashboard embedding (embed PECH-branded dashboards in customer portals) - Role-based access (different dashboards per vertical) - Apache-2.0 — can embed in proprietary products

> **Replaced Metabase** — Metabase is **AGPL-3.0**, which requires sharing source code of modifications when served to users. Since PECH will embed custom dashboards in subscription products, AGPL is incompatible.

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Metabase | **AGPL-3.0** | **Must open-source modifications when served** — incompatible with PECH's model | | Grafana | AGPL-3.0 | Same AGPL issue; better for metrics than BI | | Redash | BSD-2 | Simpler than Superset; less visualization options |

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12. AI/ML INFRASTRUCTURE

Model Serving

Tool	License	PECH Use	Link
Ollama	MIT	Phase 1 — simple model management, pull & run	github.com/ollama/ollama
vLLM	Apache-2.0	Phase 2+ — production serving with batching, high throughput	github.com/vllm-project/vllm
llama.cpp	MIT	CPU inference, GGUF quantized models	github.com/ggerganov/llama.cpp

AI Frameworks

Tool	License	PECH Use	Link
LangChain	MIT	RAG pipelines, chains, document loaders	github.com/langchain-ai/langchain
LangGraph	MIT	AI agent framework (multi-step reasoning)	github.com/langchain-ai/langgraph
Haystack	Apache-2.0	Alternative RAG framework	github.com/deepset-ai/haystack
Sentence Transformers	Apache-2.0	Embedding generation	github.com/UKPLab/sentence-transformers

ML Libraries

Tool	License	PECH Use	Link
PyTorch	BSD-3	Deep learning framework	github.com/pytorch/pytorch
Hugging Face Transformers	Apache-2.0	Model hub, tokenizers, fine-tuning	github.com/huggingface/transformers
PEFT	Apache-2.0	Parameter-efficient fine-tuning (LoRA, QLoRA)	github.com/huggingface/peft
scikit-learn	BSD-3	Traditional ML	github.com/scikit-learn/scikit-learn

Deployment Path: 1. **Phase 1:** Ollama (pull models, run with one command) 2. **Phase 2:** Ollama for dev, vLLM for production (higher throughput) 3. **Phase 3:** vLLM cluster with load balancing via APISIX

13. VECTOR DATABASE

Primary: Qdrant

Attribute	Detail
Website	qdrant.tech
GitHub	github.com/qdrant/qdrant
License	Apache-2.0
Language	Rust
PECH Use	Store embeddings for RAG pipeline, semantic search, product similarity

Why Qdrant: - Written in Rust — extremely fast and memory-efficient - Built-in filtering (combine vector search with metadata filters) - Scales from single node to cluster - REST + gRPC APIs - Apache-2.0

PECH RAG Pipeline: 1. Business data (products, docs, support tickets) → chunked 2. Chunks → embedding model (MiniLM/Nomic) → vectors 3. Vectors stored in Qdrant with metadata 4. User query → embed → Qdrant nearest-neighbor search → context 5. Context + query → LLM → answer

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Milvus | Apache-2.0 | Heavier; needs etcd + MinIO; overkill for Phase 1 | | ChromaDB | Apache-2.0 | Simpler but less production-ready than Qdrant | | Weaviate | BSD-3 | Good alternative; slightly less performance than Qdrant | | pgvector | PostgreSQL License | Viable for small scale; lacks Qdrant's advanced features |

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14. DATABASES

Primary Stack

Database	License	PECH Use	Link
PostgreSQL	PostgreSQL License (MIT-like)	Primary OLTP database for all microservices	postgresql.org
Apache AGE	Apache-2.0	Graph queries on PostgreSQL (knowledge graphs, relationships)	github.com/apache/age
Redis	BSD-3 (v7.0) / RSALv2+SSPLv1 (v7.4+)	Caching, sessions, real-time pub/sub	github.com/redis/redis
DuckDB	MIT	Analytical queries, OLAP, data exploration	github.com/duckdb/duckdb

> **Redis License Note:** Redis changed to dual RSALv2+SSPL from v7.4+. Use **Redis v7.0** (BSD-3) or switch to **Valkey** (BSD-3, Linux Foundation fork) or **KeyDB** (BSD-3) for fully permissive licensing.
>
> **Replaced Neo4j** — Neo4j Community is GPL-3.0, Enterprise is commercial. **Apache AGE** runs graph queries directly on PostgreSQL (which PECH already uses), saving an entire database deployment. Apache-2.0 licensed.

Alternatives for Graph: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Neo4j Community | GPL-3.0 | GPL; requires separate database deployment | | ArangoDB | Apache-2.0 | Full graph DB; overkill when AGE works on existing PostgreSQL |

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15. MESSAGE QUEUE / EVENT STREAMING

Primary: Apache Kafka

Attribute	Detail
Website	kafka.apache.org
GitHub	github.com/apache/kafka
License	Apache-2.0
Language	Java / Scala
PECH Use	Event streaming between microservices, real-time data pipelines, IoT data ingestion

Why Kafka: - Industry standard for event streaming - Handles millions of events per second - Durable message storage (replay capability) - Connect ecosystem for database CDC (change data capture)

Startup Alternative: NATS

Attribute	Detail
GitHub	github.com/nats-io/nats-server
License	Apache-2.0
Why Consider	Much simpler to deploy; single binary; lower resource usage; JetStream for persistence

PECH Recommendation: Start with **NATS** (Phase 1 — simpler, lighter). Migrate to **Kafka** when event volume requires it (Phase 2/3).

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16. MAPPING & GEOSPATIAL

PMAP / AMAP Platform Stack

Tool	License	PECH Use	Link
OpenStreetMap	ODbL	Base map data for Nigeria/Africa	openstreetmap.org
MapLibre GL JS	BSD-3	Map rendering frontend (replaced Mapbox GL JS)	github.com/maplibre/maplibre-gl-js
Nominatim	GPL-2.0	Geocoding (address lookup) — use as service	github.com/osm-search/Nominatim
tile-server-gl	BSD-2	Self-hosted map tile server	github.com/maptiler/tileserver-gl
PostGIS	GPL-2.0	Geospatial PostgreSQL extension — use as service	postgis.net
OSRM	BSD-2	Routing (see Logistics section)	project-osrm.org

PECH PMAP Features: - Africa-focused digital mapping platform - Business location directory - Solar installation site assessment - Delivery zone mapping - Installer coverage visualization

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17. PROCUREMENT / TENDERING

Primary: OpenProcurement

Attribute	Detail
Website	openprocurement.io
GitHub	github.com/ProzorroUKR/openprocurement.api
License	Apache-2.0
Language	Python
PECH Use	Solar project tenders, government procurement, vendor bidding, RFQ management

Why OpenProcurement: - Built for government/enterprise procurement - Transparent bidding process - Multi-lot tender support - Apache-2.0

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18. CMS / CONTENT MANAGEMENT

Primary: Strapi

Attribute	Detail
Website	strapi.io

GitHub	github.com/strapi/strapi
License	MIT (Community Edition)
Language	TypeScript (Node.js)
PECH Use	Blog, knowledge base, help center, marketing content, API documentation portal

Alternatives Considered: | Alternative | License | Why Not Chosen | |----------|---------|----------| | Directus | GPL-3.0 (core) / BSL (cloud) | GPL for core; business license issues | | Ghost | MIT | Blog-focused only; less API flexibility | | Payload CMS | MIT | Good alternative; newer; smaller community |

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19. MONITORING & OBSERVABILITY

Tool	License	PECH Use	Link
Prometheus	Apache-2.0	Metrics collection	github.com/prometheus/prometheus
Grafana	AGPL-3.0	Metrics dashboards (use as service, don't modify)	github.com/grafana/grafana
Loki	AGPL-3.0	Log aggregation (use as service)	github.com/grafana/loki
Jaeger	Apache-2.0	Distributed tracing	github.com/jaegertracing/jaeger
Uptime Kuma	MIT	Uptime monitoring	github.com/louislam/uptime-kuma

> **Grafana/Loki Note:** These are AGPL-3.0 but we use them as internal monitoring tools (not modified or embedded in products). AGPL only triggers when you modify and serve the software — using as-is for internal dashboards is fine.

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20. CONTAINER ORCHESTRATION

Tool	License	PECH Use	Link
Docker	Apache-2.0 (Engine)	Container runtime	github.com/moby/moby
Kubernetes (K8s)	Apache-2.0	Container orchestration (Phase 2+)	github.com/kubernetes/kubernetes
K3s	Apache-2.0	Lightweight Kubernetes (Phase 2)	github.com/k3s-io/k3s
Portainer	Zlib	Docker management UI	github.com/portainer/portainer

Deployment Path: 1. **Phase 1:** Docker Compose + Portainer (simple, visual management) 2. **Phase 2:** K3s (lightweight K8s for single/few nodes) 3. **Phase 3:** Full Kubernetes cluster

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21. CI/CD

Tool	License	PECH Use	Link
Gitea	MIT	Self-hosted Git (code repository)	github.com/go-gitea/gitea
Woodpecker CI	Apache-2.0	CI/CD pipelines	github.com/woodpecker-ci/woodpecker
Harbor	Apache-2.0	Container registry	github.com/goharbor/harbor

Why Not GitHub/GitLab: - GitHub Actions is proprietary (vendor lock-in) - GitLab CE is MIT but heavyweight - Gitea + Woodpecker = lightweight, self-hosted, fully open

22. LICENSE REPLACEMENTS SUMMARY

Tools originally considered that had license issues, and their PECH-approved replacements:

Original Tool	Problem License	Replacement	License	Why Better
Metabase	AGPL-3.0	**Apache Superset**	Apache-2.0	Same BI capabilities; can embed in products
n8n	Sustainable Use License	**Apache Airflow**	Apache-2.0	Full workflow automation; no resale restrictions
YOLOv8	AGPL-3.0	**RT-DETR (PaddlePaddle)**	Apache-2.0	Real-time detection; no copyleft
Sharetribe Flex	Proprietary SaaS	**Medusa.js**	MIT	Already in stack; extend for all marketplace needs
Neo4j	GPL-3.0 (Community)	**Apache AGE**	Apache-2.0	Graph queries on existing PostgreSQL; no extra DB
AutoGen	MIT (but deprecated)	**LangGraph**	MIT	More mature; better agent framework
Redis 7.4+	RSALv2+SSPL	**Redis 7.0 / Valkey**	BSD-3	Pin version or use Linux Foundation fork

23. INTEGRATION MAP

How all 25+ platforms connect in the PECH ecosystem:

```
┌──────────────────────────────────────────────────────────────┐
│                        USER REQUESTS                         │
│          (Web, Mobile, WhatsApp, IoT Devices)                │
└──────────────────────────────────────────────────────────────┘
                              │
                    ┌─────────▼─────────┐
                    │ Apache APISIX     │  ← API Gateway (rate limit, auth, routing)
                    │   (Gateway)       │
                    └─────────┬─────────┘
                              │
        ┌─────────────────────┼─────────────────────┐
        │                     │                     │
   ┌────▼────┐           ┌────▼────┐           ┌────▼────┐
   │Keycloak │           │ Medusa  │           │ Chatwoot│
   │ (Auth)  │           │(Commerce│           │(Support)│
   └─────────┘           │ + Mktpl)│           └────┬────┘
                         └────┬────┘                │
                              │                  ┌──▼──────┐
        ┌─────────────────────┼──────────────────│ AI Core │  ← LLMs via Ollama/vLLM
        │                     │                  │(Qwen,   │
   ┌────▼────┐           ┌────▼────┐           │Whisper, │
   │ ERPNext │           │Fineract │           │ etc.)   │
   │  (ERP)  │           │(Fintech)│           └────┬────┘
   └─────────┘           └─────────┘                │
                                                 ┌──▼──────┐
        ┌────────────────────────────────────────│ Qdrant  │  ← Vector DB for RAG
        │                                        │(Vectors)│
   ┌────▼────┐                                   └─────────┘
   │ThingsBoard│
   │  (IoT)  │           ┌─────────┐           ┌─────────┐
                         │  OSRM   │           │ Superset│
                         │(Routing)│           │  (BI)   │
```

```
                    └──────────┘    └──────────┘
                          │
                  ┌──────────▼──────────┐
                  │ PostgreSQL          │ ← Primary database (+ AGE for graphs)
                  │ + Redis             │ ← Cache + sessions
                  │ + NATS/Kafka        │ ← Event streaming
                  └─────────────────────┘
```

This document is confidential to PECH Group Holdings Ltd. Last updated: March 2026.